

DRAWING LIST FOR TEST PACKAGE

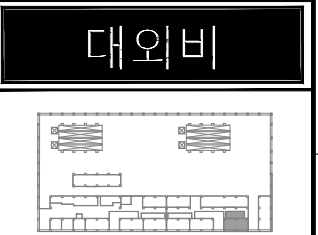
PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

PAC-510

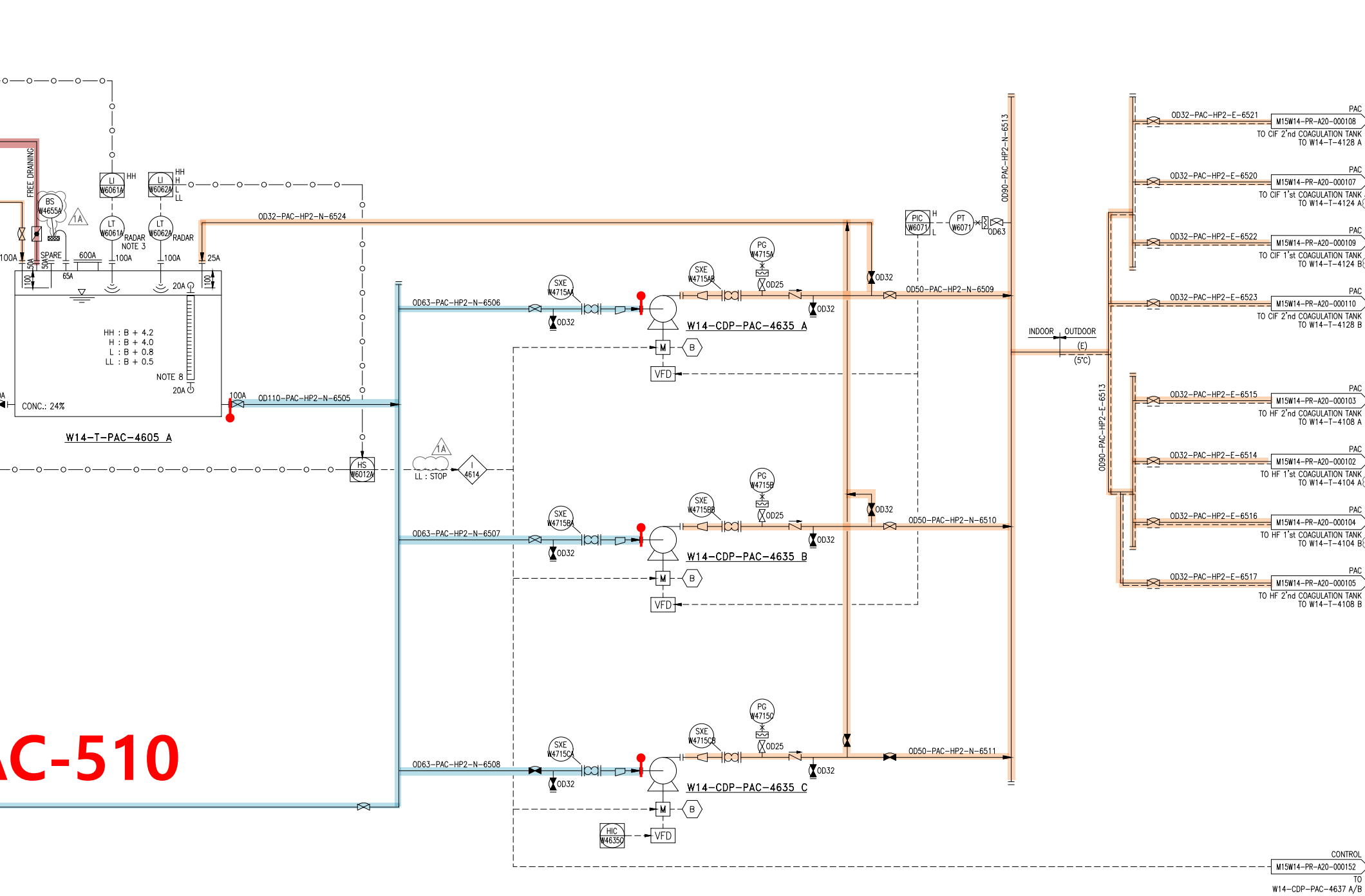
No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	510	4FA	OD63-PAC-HP2-N-6506-01	1 OF 1	0	CJ-M15W14-PR-A20-151	1A	
2	510	4FA	OD63-PAC-HP2-N-6507-01	1 OF 1	0	CJ-M15W14-PR-A20-151	1A	
3	510	4FA	OD63-PAC-HP2-N-6508-01	1 OF 1	0	CJ-M15W14-PR-A20-151	1A	
4	510	4FA	OD63-PAC-HP2-N-6532-01	1 OF 1	0	CJ-M15W14-PR-A20-000152	1A	
5	510	4FA	OD63-PAC-HP2-N-6533-01	1 OF 1	0	CJ-M15W14-PR-A20-000152	1A	
6	510	4FA	OD110-PAC-HP2-N-6505-01	1 OF 1	0	CJ-M15W14-PR-A20-151	1A	
7	510	4FA	OD110-PAC-HP2-N-6525-01	1 OF 1	0	CJ-M15W14-PR-A20-151/152	1A	
8	510	4FA	OD110-PAC-HP2-N-6531-01	1 OF 1	0	CJ-M15W14-PR-A20-000152	1A	

ITEM NO.	W14-T-PAC-4605 A		
SERVICE	PAC STORAGE TANK		
Q'TY	1		
TYPE	VERTICAL & CYLINDRICAL		
SIZE	ID x H	m	4.0 x 4.5 (EH/4.2) 1A
CAPA.	m3		57 (EH/53)
MAT'L	FRP		

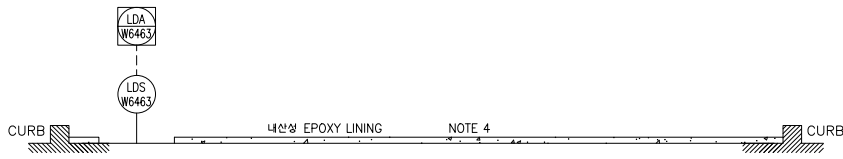


KEY PLAN (4F)

- NOTE
1. CENTRIFUGAL PUMP
MAT'L 표기 : CASING/IMPELLER
 2. 약품종류에 따라 LINE 색상을 구분하여야 한다.
 3. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
 4. CHEMICAL STORAGE TANK CURB 및 PUMP AREA에서 되는 DRAIN은 인근의 TRENCH로 접속되어, MAIN HEADER로 이동된다.
 5. INTERLOCK NO. I-4654
(W13-CTP-PAC-4674 AUTO STOP)
REMOTE STATUS에서 W14-T-PAC-4605 A/B
LEVEL (U-W6061 A/B) HIGH HIGH STOP.
 6. INTERLOCK NO. I-4614
(W14-CDP-PAC-4635 A/B/C
REMOTE STATUS에서 W14-T-PAC-4605 A/B
LEVEL (U-W6062 A/B) LOW LOW STOP.
시 CLOSE 되는 배관에는 LOTO 설치.
TANK의 역변 레벨 개시지는 VENDOR에서 공급한다.



PAC-510



ITEM NO.		W14-CDP-PAC-4635 A/B/C	
SERVICE		PAC FEED PUMP	
Q'TY		2 + 1	
TYPE		CENTRIFUGAL (VWF)	
CAPA.	m3/hr.ea x MPa	4 x 0.46	
MAT'L		FRP/FRP	
POWER	kW	{ 11 }	1A

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	실명 DESCRIPTION	날짜 DATE

제도	이인학
검토	한진희
승인	박동명
제각사	SK 에코플랜트

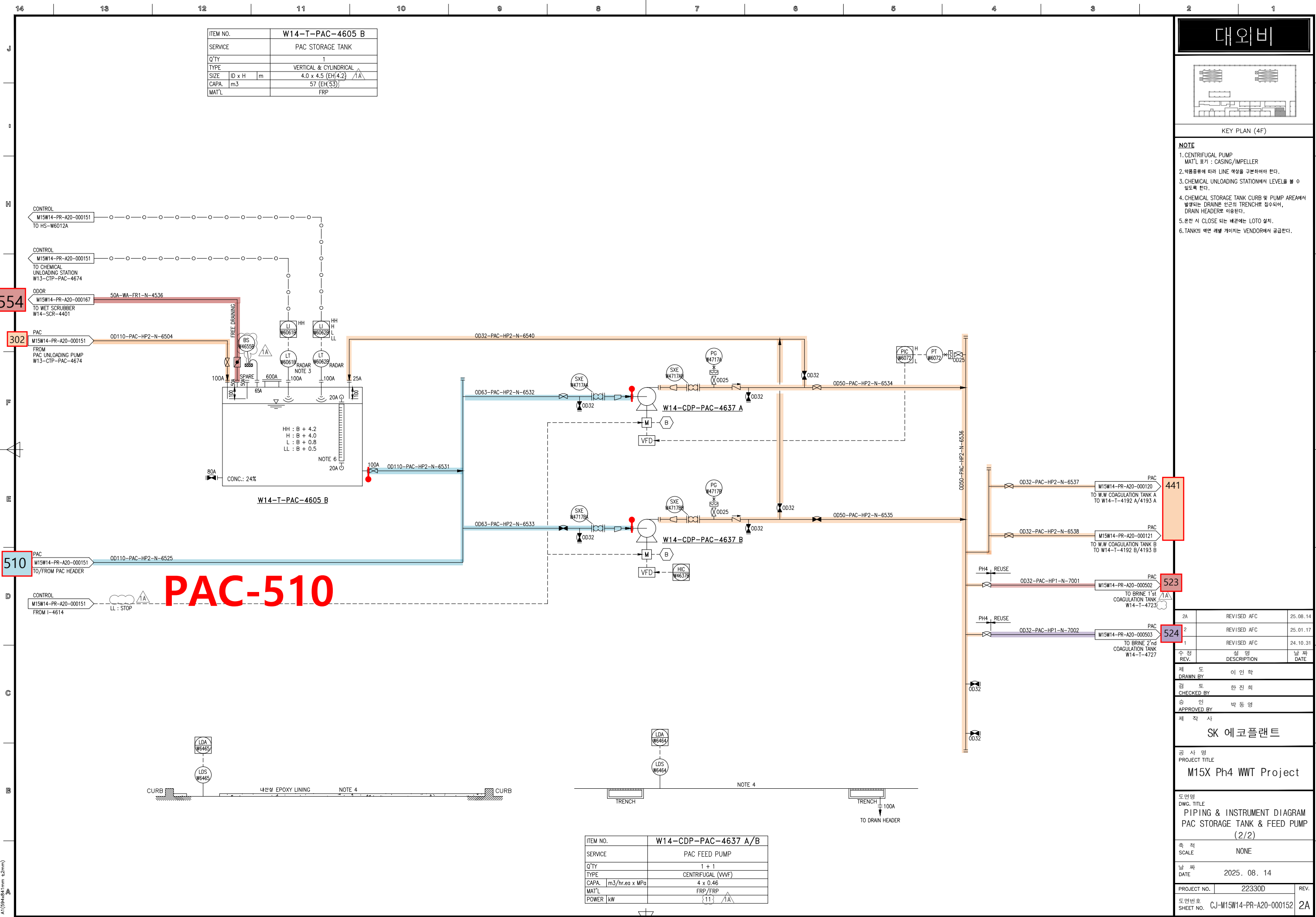
공사명
PROJECT TITLE
M15X Ph4 WWT Project

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
PAC STORAGE TANK & FEED PUMP
(1/2)

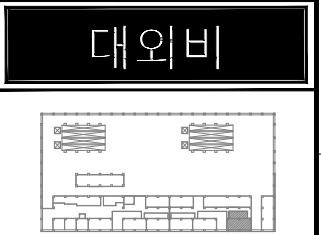
축척
SCALE
NONE

날짜
DATE
2025. 08. 14

PROJECT NO.	22330D	REV.	A
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000151	1A	



ITEM NO.	W14-T-PAC-4605 B		
SERVICE	PAC STORAGE TANK		
Q'TY	1		
TYPE	VERTICAL & CYLINDRICAL		
SIZE	ID x H	m	4.0 x 4.5 (EH/4.2) / 1A
CAPA.	m3		57 (EH/53)
MAT'L	FRP		



- NOTE
1. CENTRIFUGAL PUMP
MAT'L 표기 : CASING/IMPELLER
 - 2.약품종류에 따라 LINE 색상을 구분하여야 한다.
 3. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
 4. CHEMICAL STORAGE TANK CURB 및 PUMP AREA에서 발생되는 DRAIN은 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이송된다.
 5. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
 6. TANK의 역전 레벨 계어지는 VENDOR에서 공급한다.

PAC-510

ITEM NO.	W14-CDP-PAC-4637 A/B		
SERVICE	PAC FEED PUMP		
Q'TY	1 + 1		
TYPE	CENTRIFUGAL (VVF)		
CAPA.	m3/hr.ea x MPa		4 x 0.46
MAT'L	FRP/FRP		
POWER kW	{11} / 1A		

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정 REV.	실명 DESCRIPTION	날짜 DATE
제도	이인학	
검토	한진희	
승인	박동명	
제작사	SK 에코플랜트	
공사명 PROJECT TITLE	M15X Ph4 WWT Project	
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM PAC STORAGE TANK & FEED PUMP (2/2)	
축척 SCALE	NONE	
날짜 DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000152	2A



1

CONTRACTOR 

<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	—	DESIGN	1	60	UNIT 4FA						
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A						
						REF. P & ID CJ-M15W14-PR-A20-000151		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동호
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'

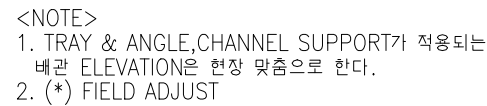


2

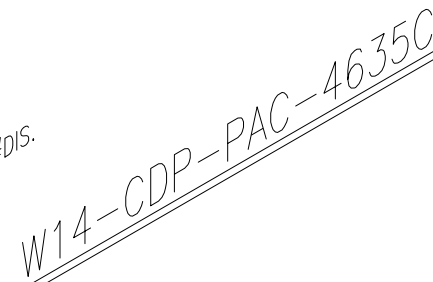
CONTRACTOR 

COMMODITY	PAC
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LINE NO. OD63-PAC-HP2-N-6507



INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	-	DESIGN	1	60	UNIT 4FA						
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A						
						REF. P & ID CJ-M15W14-PR-A20-000151		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP



1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

CLIENT



3

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1	60	UNIT	4FA							M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.							COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040013-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.
						CJ-M15W14-PR-A20-000151								0D63-PAC-HP2-N-6508	01 / 01
															0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음

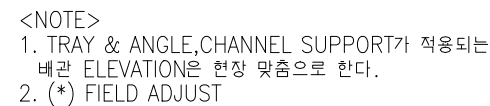


4

CONTRACTOR 

COMMODITY	PAC
ALUMINUM	
COPPER	
GOLD	
SILVER	
WHEAT	
CORN	
BUSINESS TRAVEL	
RENTS	
TELEPHONE	
TRAVEL EXPENSES	
UNEMPLOYMENT COMPENSATION	
VOLUNTARY CONTRIBUTIONS	
WAGES AND SALARIES	
WORKERS' COMPENSATION	
OTHER	
TOTAL	

LINE NO.	0007	PLC	W20	N
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* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



5

CONTRACTOR 

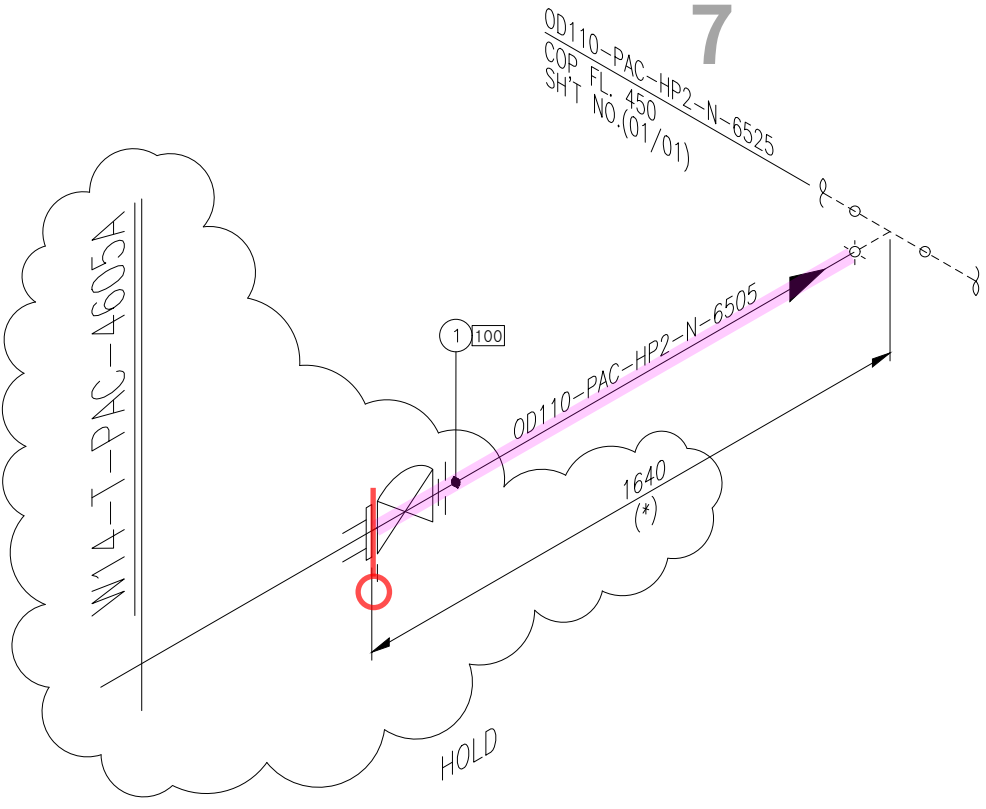
<NOTE>
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배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB20	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1	60	UNIT	4FA							M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG.							COMMODITY	JOB NO.		
						CJ-M15W14-PI-A30-040014-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	PAC	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000152								0D63-PAC-HP2-N-6533	01 / 01	0



ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD110	1.3M	PIPE, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
VD	OD110	1EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY--PP /
			TRIM--EPDM+PTFE DIAPHRAGM, BB, HANDWHEEL
F	OD110	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
			LAP JOINT--STS304 / STUB END--HDPE PE100
G	OD110	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B1	OD110	2SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
SH	OD110	2EA	SHIELD



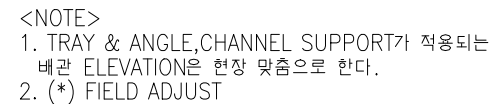
<NOTE>
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2. (*) FIELD ADJUST

CLIENT



6

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1	60	UNIT	4FA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A								COMMODITY	PAC	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000151			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	0D110-PAC-HP2-N-6505	SH'T NO. 01 / 01	REV. 0
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				



CLIENT		7	
			
CONTRACTOR		PROJECT NAME	
COMMODITY		M15X Ph4 Project	
PAC		JOB NO.	
		22330D	
LINE NO.		SH'T NO.	REV.
OD110-PAC-HP2-N-6525		01 / 01	0

[illegible]

8

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1O/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1	60	UNIT	4FA							M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040014-A						COMMODITY		PAC	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000152		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	OD110-PAC-HP2-N-6531	SH'T NO. 01 / 01	REV. 0
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				